

Arduino-Based Capacitance Measurement – Early POC for Smart Tampon

Objective

The purpose of this system is to measure the capacitance of a tampon using an Arduino-based setup and to transmit the data to a computer for analysis. In subsequent experiments, specific capacitance values will be mapped to saturation levels, where **10 mL of absorbed fluid will represent 100% saturation**.

System Overview

The measurement system is based on a modified version of the Arduino capacitance meter described in the following guide:

 [Measure Capacitance with Arduino – Instructables](#)

This system allows the estimation of low capacitance values in the picofarad (pF) range, suitable for detecting small changes in the tampon's capacitance due to fluid absorption.

Components

Component	Quantity	Description
Arduino Uno (or Nano)	1	Main microcontroller unit
1 MΩ Resistor	1	Used to form RC timing circuit
Test capacitor (tampon)	1	Two foil strips embedded along tampon sides
Breadboard + jump wires	–	For prototyping connections

USB cable	1	For Arduino-PC communication
Optional: PC UI interface	1	(e.g. Serial Monitor or Python GUI)

Measurement Principle

The Arduino code measures how long it takes for a known resistor (1 MΩ) to charge the unknown capacitor (i.e., the tampon) up to a threshold voltage. The time duration is proportional to the capacitance value.

Formula used:

Time to threshold $\approx R \times C \times \ln(V_{\text{final}}/V_{\text{initial}})$

The code then translates the time into an estimated capacitance value and sends it via serial communication to the connected computer.

System Construction Steps

1. Tampon Capacitor Preparation:

- Attach two identical aluminum foil strips longitudinally along the tampon, acting as the capacitor plates.
- Ensure strips do not touch each other.
- Optionally insulate with a thin dielectric layer (e.g., plastic wrap or tape).

2. Arduino Circuit Setup:

- Connect one foil strip to a digital pin on the Arduino (e.g., D8).
- Connect the second foil strip to GND.
- Connect a 1 MΩ resistor between the digital pin and another pin configured for timing measurements (e.g., D9).

3. Upload Arduino Code:

- Use the code provided in the Instructables guide to calculate capacitance based on timing.
- Modify the code if needed to improve sensitivity in the range of 5–50 pF.

4. **Connect to PC:**

- Use the Serial Monitor or a custom Python-based GUI to read and log capacitance data in real time.

Next Phase – Calibration

In future experiments:

- A **synthetic fluid simulating menstrual blood** will be gradually added to the tampon to observe capacitance changes at increasing saturation levels.
- The composition of the simulated fluid is as follows:

Simulated Menstrual Fluid Composition

- 100 mL warm water (non-boiling)
- 0.9 g table salt
- 30 mL honey
- 0.2 g cornstarch
- Red food coloring (for visual reference)
- The fluid will be applied incrementally using a syringe.
- Capacitance readings will be taken after each fluid increment.
- A mapping table will be constructed to correlate capacitance values to tampon saturation percentages:
 - **0% saturation** = dry tampon
 - **100% saturation** = 10 mL absorbed fluid
 - Intermediate levels will be estimated based on the data curve.

Planned Enhancements

- Improved signal stability and noise reduction (e.g., averaging multiple readings).
- Integration with BLE module for wireless data transfer.
- Encapsulation of electrodes for medical-grade safety in future prototypes.

Conclusion

This preliminary setup provides a low-cost and effective method for measuring the capacitance of a tampon and transmitting the results to a computer. It forms the basis for developing a real-time fluid monitoring system that will later be enhanced with calibrated saturation-level detection and potential mobile integration.